Lab Note 7: Estimating Energy from Mass

This note shows a short program that applies the mass-energy relation E = mc^2 to a list of masses. Here E is the energy in joules, m is the mass in kilograms and c is the speed of light.

```
C = 299792458  # speed of light in m/s

def energy(mass_kg):
    """Return the rest energy in joules."""
    return mass_kg * C ** 2

for m in [0.001, 0.5, 2.0]:
    print(f"m = {m} kg -> E = {energy(m):.3e} J")
```

The program prints one line for each mass in the list. Results are written in scientific notation with three decimal places.

The ratio of two energies is simply E1/E2 = m1/m2, because the constant c^2 cancels out. For example, doubling the mass doubles the energy.